Quantification: 876_z89c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	10
interior lacuna count	9
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2175.11	2169	869.225	px^2
major_axis_length_px	lacuna shape	84.7631	81.6741	19.9666	px
minor_axis_length_px	lacuna shape	33.7022	35.8518	7.7268	px
aspect_ratio	lacuna shape	2.557	2.7987	0.4604	unitless
eccentricity	lacuna shape	0.9107	0.934	0.04	unitless
solidity	lacuna shape	0.8171	0.8169	0.0544	unitless
roots_count	headline	12.5556	12	5.1747	count
ring_length_r30_px	headline	725.444	724	105.746	px
ring_length_r60_px	reported	1917.89	2031	368.944	px
owned_length_px	ownership-dependent	6802.65	7507.19	2709.44	px
edge_count	ownership-dependent	505.556	575	199.205	count
mean_edge_length_px	ownership-dependent	13.4453	13.3372	0.4119	px
perimeter_px	normalised	294.887	274.676	88.6524	px
ring_area_r30_px2	normalised	9333.67	9180	1991.24	px^2
ring_area_r60_px2	normalised	23113.9	23485	4641.06	px^2
in_frame_fraction_r30	normalised	0.9828	1	0.0515	unitless
in_frame_fraction_r60	normalised	0.9539	1	0.0857	unitless
ring_density_r30	normalised	0.0793487	0.0796067	0.0127441	px^-1
ring_density_r60	normalised	0.0835927	0.0821726	0.0114971	px^-1
roots_per_100px_perimeter	normalised	4.2308	4.2819	1.1123	per 100 px
ring_attached_length_r30_px	network v2	654.333	700	158.803	px
ring_attached_length_r60_px	network v2	1731.78	1711	497.802	px
ring_length_w_r30_px	network v2	817.611	824.671	120.745	px
ring_length_w_r60_px	network v2	2173.85	2323.93	422.33	px
sholl_crossings_r10	network v2	16.1111	17	3.4075	count
sholl_crossings_r20	network v2	19.5556	20	4.1265	count
sholl_crossings_r30	network v2	22.5556	22	4.8762	count
loop_count	loops, report only	83.4444	98	37.9806	count
loop_edge_share	loops, report only	0.5707	0.5663	0.0929	share
ring_width_mean_r30_px	width	5.9702	5.6859	0.7533	px

Per field
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measure	value	units
analysed_area_px2	1.02251e+06	px^2
total_skeleton_length_px	77353	px
canalicular_length_density_per_px	0.0756501	px^-1
junction_count	2863	count
junction_density_per_px2	0.00279997	px^-2
lacuna_count	10	count

lacunae_per_px2	9.78e-06	px^-2
skeleton_component_count	542	count
mean_component_length_px	142.718	px
median_component_length_px	10	px
field_length_density_w_per_px	0.0852398	px^-1
field_density_without_flagged_per_px	0.0765378	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	90	px
unattached_component_count	6	count
canalicular_length_density_attached_per_px	0.0756501	px^-1
field_length_density_with_unattached_per_px	0.0757381	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	969	count
loop_edge_share	0.5389	share
edge_count	6066	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.297	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	62911	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	6490	249.288	41.8559	5.9559	0.9858	0.5841	-1.5551	22.8282	886.791	True
2	3313	115.919	39.5634	2.93	0.94	0.7245	-1.5598	154.062	372.366	False
3	2754	105.554	35.8518	2.9442	0.9406	0.7558	1.5255	165.057	102.925	False
4	3301	102.585	44.3824	2.3114	0.9016	0.8169	-1.4051	307.062	909.902	False
5	2169	78.4025	38.343	2.0448	0.8723	0.8413	1.557	311.008	321.631	False
6	956	61.2609	21.1134	2.9015	0.9387	0.8	1.5273	461.978	61.1475	False
7	2610	87.8807	38.6915	2.2713	0.8979	0.8782	-1.531	535.001	198.764	False
8	1500	59.3512	33.3189	1.7813	0.8276	0.8966	1.5421	545.937	440.713	False
9	1670	81.6741	26.9572	3.0298	0.944	0.8337	-1.4297	692.65	612.224	False
10	1303	70.2403	25.0978	2.7987	0.934	0.8068	-1.3129	914.738	39.4635	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	34	1057	2018	7840.01	569	13.7786	
2	23	870	2363	9747.15	704	13.8454	5.2883
3	15	778	1852	4252.91	321	13.2489	5.6859
4	16	705	2099	9589.3	699	13.7186	5.4671
5	12	807	2098	5321.55	399	13.3372	6.7128
6	12	724	1641	3740.62	290	12.8987	6.9903
7	13	753	2188	7507.19	575	13.056	6.7179

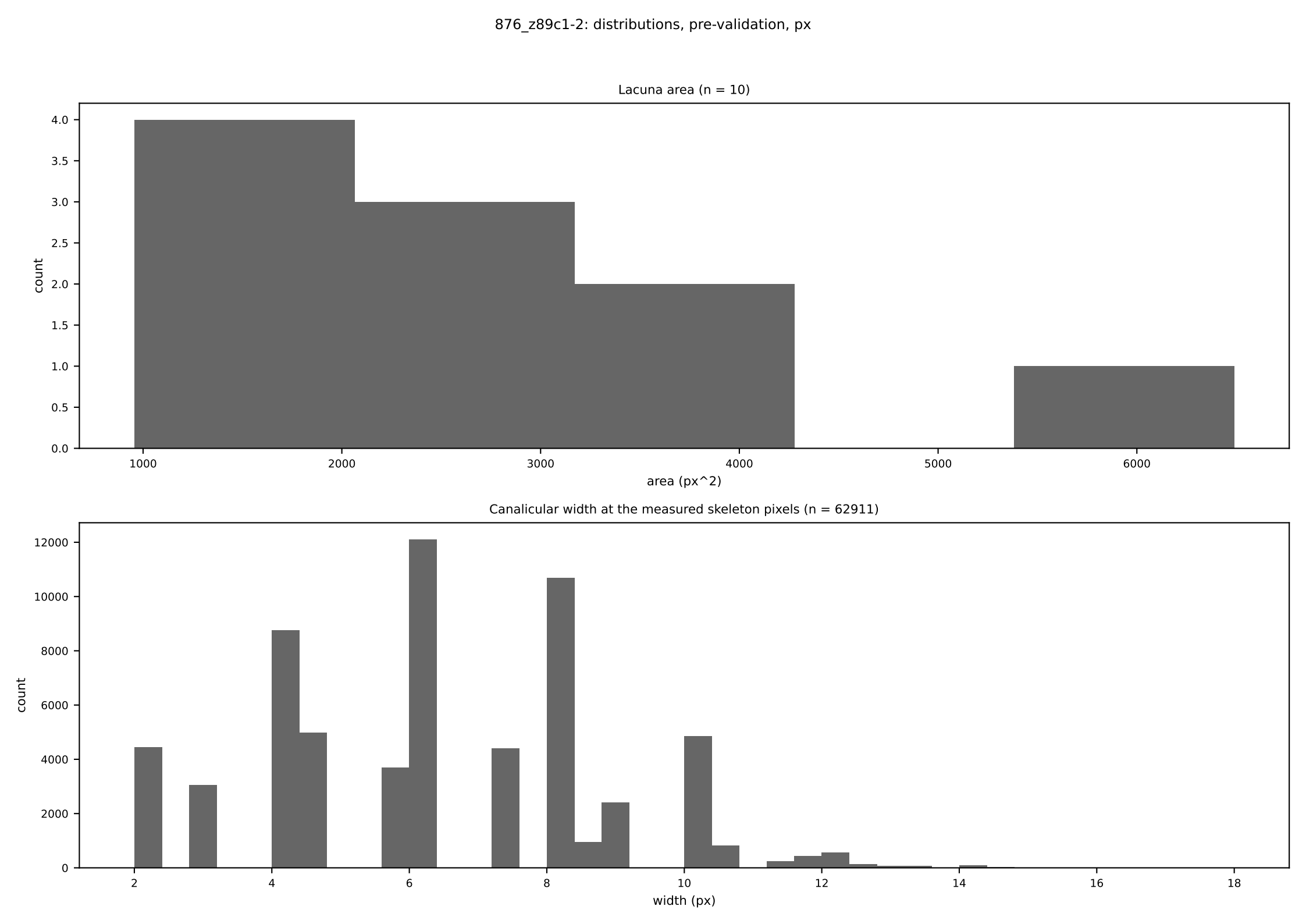
8	6	708	2031	9832.18	753	13.0573	6.3834
9	9	697	1882	8038.6	581	13.8358	5.6684
10	7	487	1107	3194.32	228	14.0102	4.818

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	1053.68	14114	25792	0.6408	0.5406	0.0748902	0.0782413	3.2268
2	423.014	12381	29774	1	1	0.070269	0.0793646	5.4372
3	413.328	11320	26755	1	0.9628	0.0687279	0.0692207	3.6291
4	373.665	11365	27711	1	0.9875	0.0620326	0.0757461	4.2819
5	273.505	9180	23731	1	1	0.0879085	0.0884076	4.3875
6	191.315	7221	16849	1	0.887	0.100263	0.0973945	6.2724
7	277.848	9459	23485	1	1	0.0796067	0.0931658	4.6788
8	190.593	7507	20585	1	1	0.094312	0.0986641	3.1481
9	274.676	8714	22903	1	1	0.0799862	0.0821726	3.2766
10	236.043	6856	16232	0.8455	0.748	0.0710327	0.0681986	2.9656

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	1037	1910	1178.44	2232.76	29	19	22
2	797	2280	966.482	2647.2	19	22	28
3	720	1660	874.826	2104.34	21	20	22
4	532	1693	785.37	2373	14	22	26
5	781	1922	915.683	2378.7	18	20	22
6	685	1550	824.671	1845	18	18	22
7	731	2182	861.754	2505.51	15	20	26
8	700	2008	806.399	2323.93	17	17	22
9	659	1711	783.671	2148.56	13	26	24
10	284	580	539.647	1238.39	10	11	11


876_z89c1-2: distributions, pre-validation, px
Lacuna area (n = 10)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 62911)
count
width (px)